

LAYER 3

The Folding Operation (Δ)

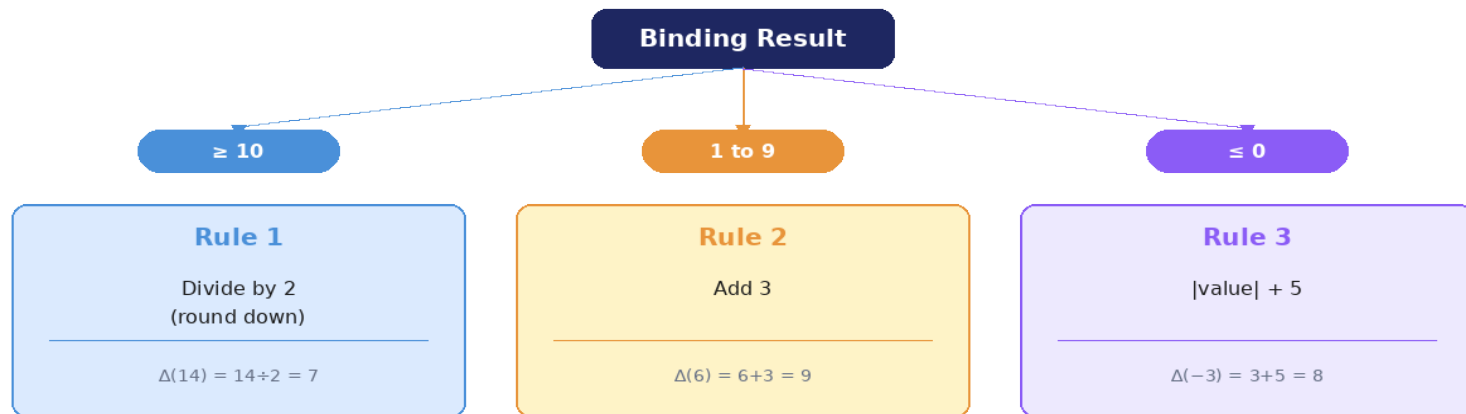
Three conditional rules that transform any binding result into a positive number.

What Is Folding?

After Binding two elements, you often need to Fold the result. Folding takes any number — positive, negative, or zero — and transforms it using one of three conditional rules.

Which rule applies depends entirely on the size and sign of the input:

1. Result is 10 or greater → Rule 1
2. Result is positive but less than 10 (1 through 9) → Rule 2
3. Result is zero or negative → Rule 3



Rule 1: Result $\geq 10 \rightarrow$ Divide by 2, Round Down

When the binding result is 10 or greater, divide it by 2. If the answer is not a whole number, round down.

$$\Delta(14) = 14 \div 2 = 7$$

$$\Delta(10) = 10 \div 2 = 5$$

$$\Delta(19) = 19 \div 2 = 9.5 \rightarrow \text{rounded down} = 9$$

$$\Delta(100) = 100 \div 2 = 50$$

This rule always makes the number smaller. A large result gets compressed.

Practice: $\Delta(17) = ?$ and $\Delta(25) = ?$

Answers: $\Delta(17) = 8$ (rounded down from 8.5) | $\Delta(25) = 12$

Rule 2: Result 1–9 → Add 3

When the binding result is positive but less than 10 (the numbers 1 through 9), simply add 3.

$$\Delta(1) = 1 + 3 = 4$$

$$\Delta(6) = 6 + 3 = 9$$

$$\Delta(9) = 9 + 3 = 12 \quad \leftarrow \text{output can exceed 10!}$$

 The rules check the INPUT, not the output. $\Delta(9)$ uses Rule 2 because the input is 9 (less than 10), even though the output is 12.

Practice: $\Delta(5) = ?$ and $\Delta(2) = ?$

Answers: $\Delta(5) = 8$ | $\Delta(2) = 5$

Rule 3: $\text{Result} \leq 0 \rightarrow |\text{value}| + 5$

When the result is zero or negative, take the absolute value (make it positive) and add 5.

$$\Delta(0) = |0| + 5 = 5$$

$$\Delta(-3) = |-3| + 5 = 3 + 5 = 8$$

$$\Delta(-10) = |-10| + 5 = 10 + 5 = 15$$

This rule rescues negative numbers — it always produces something positive and adds a buffer of 5.

Practice: $\Delta(-6) = ?$

$$\text{Answer: } |-6| + 5 = 6 + 5 = 11$$



Folding ALWAYS produces a positive result. No matter what goes in, the output is always positive.

Layer 3 Practice — Binding + Folding Combined

Now let's combine Layers 2 and 3. Compute the binding, then fold the result.

1. $\text{Glyph}_1 \oplus \text{Ring}_4 = ?$ then $\Delta(\text{result}) = ?$

$\text{Glyph} \oplus \text{Ring} = \text{multiply: } 1 \times 8 = 8$. Then $\Delta(8) = 8 + 3 = 11$ (Rule 2: positive < 10)

2. $\text{Ring}_1 \oplus \text{Shard}_2 = ?$ then $\Delta(\text{result}) = ?$

$\text{Ring} \oplus \text{Shard} = \text{R} \times 2 - \text{S: } (2 \times 2) - 4 = 0$. Then $\Delta(0) = |0| + 5 = 5$ (Rule 3: zero)

3. $\text{Glyph}_6 \oplus \text{Shard}_2 = ?$ then $\Delta(\text{result}) = ?$

$\text{Glyph} \oplus \text{Shard} = \text{S} - \text{G: } 4 - 6 = -2$. Then $\Delta(-2) = |-2| + 5 = 7$ (Rule 3: negative)